

Carbon Footprint Report

Apex Manufacturing

GHG Emissions Report — 2024

Prepared: July 06, 2026

Industry Sector	Technology
Country	United States
Number of Employees	250
Annual Revenue	\$5,000,000
Reporting Year	2024

Emission Summary

43.86 tCO ₂ e Scope 1	99.17 tCO ₂ e Scope 2	295.71 tCO ₂ e Scope 3	438.74 tCO ₂ e TOTAL
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Sustainability Score: 82.5/100 — Excellent

1. Executive Summary

Apex Manufacturing has completed a comprehensive greenhouse gas (GHG) emissions inventory for the 2024 year, demonstrating our commitment to environmental transparency and sustainability. As a Technology organization with 250 employees, we recognize the critical importance of measuring and managing our carbon footprint.

Our total GHG emissions for this reporting period amount to 438.74 tonnes of CO2 equivalent (tCO2e). This comprises 43.86 tCO2e from Scope 1 direct emissions, 99.17 tCO2e from Scope 2 energy-related emissions, and 295.71 tCO2e from Scope 3 value chain emissions. Understanding these emission sources enables us to develop targeted reduction strategies.

Moving forward, we are committed to implementing science-based targets aligned with the Paris Agreement goals. Our roadmap includes transitioning to renewable energy, optimizing operational efficiency, and engaging our supply chain partners in emission reduction initiatives. We aim to achieve significant reductions in our carbon footprint while maintaining business growth and operational excellence.

2. Scope Analysis

Scope 1 (Direct Emissions): 43.86 tCO2e Scope 1 emissions represent 10.0% of our total footprint. These direct emissions come from sources owned or controlled by the organization. Key contributors include Diesel: 6700 kg, Petrol: 3465 kg, LPG: 453 kg. Reduction opportunities include fleet electrification, fuel switching, and improved maintenance protocols.

Scope 2 (Energy Emissions): 99.17 tCO2e Scope 2 accounts for 22.6% of total emissions, primarily from Electricity: 98400 kg, Steam: 540 kg, Cooling: 225 kg. This represents indirect emissions from the generation of purchased energy. The most impactful reduction strategy is transitioning to renewable energy through power purchase agreements (PPAs) or renewable energy certificates (RECs).

Scope 3 (Value Chain Emissions): 295.71 tCO2e Scope 3 represents 67.4% of our carbon footprint, encompassing Commuting: 13600 kg, Flight: 30600 kg, Train: 615 kg. These value chain emissions require collaborative approaches with suppliers and stakeholders. Priority actions include supplier engagement programs, travel reduction policies, and circular economy initiatives.

Scope 1 — Direct Emissions

Category	Activity	EF (kg/unit)	Emissions (kg CO2e)	Emissions (tCO2e)
Diesel	2500.00	2.6800	6700.00	6.7000
Petrol	1500.00	2.3100	3465.00	3.4650
LPG	300.00	1.5100	453.00	0.4530
NaturalGas	800.00	2.0200	1616.00	1.6160
CompanyVehicle	45000.00	0.2100	9450.00	9.4500
Refrigerant	12.50	1774.0000	22175.00	22.1750

Scope 2 — Energy Emissions

Category	Activity	EF (kg/unit)	Emissions (kg CO ₂ e)	Emissions (tCO ₂ e)
Electricity	120000.00	0.8200	98400.00	98.4000
Steam	2000.00	0.2700	540.00	0.5400
Cooling	1500.00	0.1500	225.00	0.2250

Scope 3 — Value Chain Emissions

Category	Activity	EF (kg/unit)	Emissions (kg CO ₂ e)	Emissions (tCO ₂ e)
Commuting	80000.00	0.1700	13600.00	13.6000
Flight	120000.00	0.2550	30600.00	30.6000
Train	15000.00	0.0410	615.00	0.6150
PurchasedGoods	800000.00	0.3000	240000.00	240.0000
Waste	5000.00	0.5700	2850.00	2.8500
Water	2500.00	0.3400	850.00	0.8500
Logistics	60000.00	0.1200	7200.00	7.2000

3. Risk Assessment

Regulatory Risks: The Technology sector faces increasing regulatory pressure from climate disclosure requirements including CSRD, TCFD, and potential carbon pricing mechanisms. With a current sustainability score of 82.5/100, there is significant exposure to compliance costs and reporting obligations. Organizations must prepare for mandatory climate disclosures and potential carbon taxes.

Physical Climate Risks: Climate change poses direct operational risks through extreme weather events, supply chain disruptions, and resource scarcity. The United States region may experience increased frequency of climate-related disruptions affecting business continuity. Adaptation strategies should include climate risk assessments, resilient infrastructure investments, and supply chain diversification.

Transition and Reputational Risks: Stakeholder expectations for climate action are rapidly evolving. Investors, customers, and employees increasingly prioritize environmental performance. With total emissions of 438.74 tCO₂e, demonstrating credible reduction pathways is essential for maintaining competitive advantage and stakeholder trust. Failure to act may result in loss of market share, difficulty attracting talent, and reduced access to capital.

4. SBTi-Aligned Recommendations

#1 Switch to 100% Renewable Electricity

Priority	High	Expected Reduction	20%
Cost Impact	Medium Investment	Difficulty	Moderate
Benefit	Immediate elimination of Scope 2 emissions aligned with SBTi 1.5C pathway.		

Transition all purchased electricity to certified renewable sources via PPAs or RECs. This directly eliminates Scope 2 emissions.

#2 Implement Remote Work and Virtual Meeting Policy

Priority	Medium	Expected Reduction	12%
Cost Impact	Low Cost	Difficulty	Easy
Benefit	Reduces Scope 3 travel emissions while improving employee work-life balance.		

Reduce business travel and commuting by 30% through remote work and virtual tools. Target flights first as highest-impact source.

#3 Launch Circular Economy Waste Program

Priority	Medium	Expected Reduction	8%
Cost Impact	Low Cost	Difficulty	Easy
Benefit	Reduces Scope 3 waste emissions and demonstrates circular economy leadership.		

Implement waste segregation and recycling targeting 50% landfill diversion. Partner with certified waste management providers.

5. Conclusion

This carbon footprint assessment reveals a total emission footprint of 438.74 tCO₂e with a sustainability score of 82.5/100 (Excellent). The detailed analysis across all three scopes provides a comprehensive baseline for developing targeted reduction strategies. By understanding our emission sources and implementing the recommended actions, Apex Manufacturing can make meaningful progress toward climate goals.

The path to Net Zero requires commitment, innovation, and collaboration across all levels of the organization. By aligning with Science Based Targets initiative (SBTi) methodologies and implementing the recommendations outlined in this report, we can achieve significant emission reductions while maintaining operational excellence. The time to act is now—every tonne of CO₂e reduced contributes to a more sustainable future for our planet, our stakeholders, and future generations.

This report was generated using the Carbon Footprint Report Generator for SMEs, powered by the GHG Protocol methodology and Groq AI.